THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A red die and a blue die are rolled together. The probability that the sum is a multiple of 4 is: 1

(a) $\frac{5}{18}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{9}$

2. A coin is tossed three times. The probability of getting at least one tail is: 1

(a) 1 **(b)** $\frac{7}{8}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{8}$

3. 30 cards numbered 1 to 30 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 3 or 5 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{7}{15}$ **(c)** $\frac{13}{30}$ **(d)** $\frac{8}{15}$

4. A bag contains 3 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. The value of x is: 1

(a) 9 **(b)** 6 **(c)** 2 **(d)** 5

5. A bag contains 3 red and 5 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. The value of x is: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

6. A red die and a blue die are rolled together. The probability that both numbers are even is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{5}{18}$ **(d)** $\frac{1}{4}$

7. A letter is chosen at random from the word 'PERMUTATION'. The probability that it is a vowel is: 1

(a) $\frac{4}{11}$ **(b)** $\frac{6}{11}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{5}{11}$

8. A red die and a blue die are rolled together. The probability that the first number is greater than the second is: 1

(a) $\frac{7}{12}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{7}{18}$ **(d)** $\frac{4}{9}$

9. A coin is tossed three times. The probability of getting a head on the first toss is: 1

(a) $\frac{3}{8}$ **(b)** $\frac{4}{9}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{1}{2}$

10. 40 cards numbered 1 to 40 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a composite number is: 1

(a) $\frac{7}{10}$ **(b)** $\frac{27}{40}$ **(c)** $\frac{13}{40}$ **(d)** $\frac{13}{20}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A red die and a blue die are rolled together. Find the probability that the sum is a multiple of 3. 2

12. 36 cards numbered 1 to 36 are placed in a box, mixed well and one is drawn at random. Find the probability that the number drawn is divisible by both 2 and 3. 2

13. A bag contains 6 red, 4 blue, 5 green balls. One ball is drawn at random. Find the probability that it is not green. 2

14. A red die and a blue die are rolled together. Find the probability that one number is exactly double the other. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains 7 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. Find the value of x. 3

16. A bag contains 3 red and 6 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. Find the value of x. 3

17. A red die and a blue die are rolled together. Find the probability that the sum of the numbers is 8. 3

OR

A bag contains 8 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. Find the value of x. 3

18. A bag contains 5 red and 5 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{2}{3}$. Find the value of x. 3

OR

A red die and a blue die are rolled together. Find the probability that the sum of the numbers is 7. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the product of the numbers is 12,
(ii) the sum is a prime number,
(iii) the sum is a multiple of 3 (in this order).

OR

A box contains 40 cards numbered 1 to 40. One card is drawn at random. Find the probability that the number is: 5
(i) a multiple of 3,
(ii) a prime number,
(iii) a multiple of 4 (in this order).

20. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) one number is exactly double the other,
(ii) the product is less than 8,
(iii) the larger number is 4 (in this order).

OR

A box contains 50 cards numbered 1 to 50. One card is drawn at random. Find the probability that the number is: 5
(i) a perfect square,
(ii) a number divisible by 5,
(iii) an even prime number (in this order).

END OF QUESTION PAPER — Total: 40 Marks